Vinnakota Hema Kasi Sainath

AI Engineer • Python • Generative AI

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PROFESSIONAL SUMMARY

AI Engineer with 3.5 years of experience building production-grade Generative AI systems — from end-to-end RAG pipelines and enterprise AI agents to real-time voice agents and secure multi-agent orchestration. Expertise spans AWS Bedrock, Agno, LangChain, MCP, and LLM integrations, with a strong foundation in Python backend engineering (FastAPI, Docker, Celery). Known for rapidly converting PoCs into robust, scalable solutions and enterprise-grade security.

TECHNICAL SKILLS

AI / ML: RAG Pipelines, Agentic AI, Multi-Agent Orchestration, LangChain, Agno, RAGAS, MCP (Model Context Protocol), A2A, LLMs (Claude, Gemini, OpenAI, GPT-4), Voice Agents (Pipecat, Deepgram, Cartesia, Sarvam), AWS Bedrock, AgentCore, Strands

Backend: Python (Async/Sync), FastAPI, Celery, Docker, Kafka, SQLAlchemy, SQLModel, Pydantic, asyncio, Pytest

Databases: PostgreSQL, MySQL, Neo4j, Vector DB, Supabase, OpenSearch Serverless

AI Dev Tools: Streamlit, Gradio, Cursor, Windsurf, Kiro, Postman, PgAdmin, DBeaver, Git, JIRA

Security & Reliability: JWT, RBAC, LLM Guardrails, Prompt Injection Defence, Input Sanitisation, Rate Limiting, Keycloak, Observability & Logging

PROFESSIONAL EXPERIENCE

Software Engineer I | **Smart IMS (Hyderabad)** *Dec 2022 – Present*

- Built multiple end-to-end GenAI solutions — from requirement gathering and LLM selection through RAG pipeline design, agent orchestration, evaluation (RAGAS), and production deployment with CI/CD.
- Designed and implemented RAG systems: document ingestion, chunking strategies, embedding, vector store retrieval, reranking, and RAGAS-based evaluation loops to iteratively improve accuracy.
- Engineered enterprise AI agents using Agno and AWS (Bedrock, AgentCore, Strands) with tool use, guardrails, prompt injection defence, RBAC, and JWT-secured APIs for secure multi-user access.
- Built real-time voice agents using Pipecat with Deepgram & Cartesia (English) and Sarvam (Indian regional languages) for STT/TTS over WebSockets and WebRTC — full conversational audio.
- Diagnosed and resolved critical RAG failure where retrieval quality degraded silently — root-caused to chunk-size mismatch and embedding drift; introduced RAGAS monitoring
- Used Celery and FastAPI Background Tasks for heavy, long-running AI/ML workflows; implemented structured logging and observability for production incident response.
- Collaborated in Agile sprints, translating business requirements into AI-driven solutions while maintaining alignment with engineering best practices.

PROJECTS

AI Agent for Insurance Carrier – Coaction (AWS-Native)

- Built end-to-end: scoped requirements with stakeholders, designed RAG architecture, implemented agent tools, deployed on AWS, and delivered a Streamlit interface
- RAG design: ingested SharePoint documents and web-scraped carrier data (Crawl4AI); applied semantic chunking, embedding into OpenSearch Serverless, and reranking for high-precision insurance Q&A.
- Security: enforced RBAC and JWT at the API layer; added LLM guardrails and prompt injection filters to prevent data leakage and adversarial inputs in an enterprise context.
- Containerized all services with Docker, pipelines using Pytest suites
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Tech Stack: *AWS Bedrock, AgentCore, Strands, RAG, Crawl4AI, S3, Lambda, Python, FastAPI, Vector DB, OpenSearch Serverless, Docker, Streamlit*

Xymphony AI – Agent Development Platform

- Developed a scalable backend using FastAPI and Python integrating multiple LLM providers (OpenAI, Google Gemini, Anthropic Claude) via a unified API interface.
- Built modular agent architecture with the Agno framework, supporting custom tools and dynamic RAG knowledge retrieval — enabling rapid PoC-to-production cycles.
- Implemented a RAG evaluation system using RAGAS and custom metrics to continuously improve agent accuracy and retrieval quality across diverse use cases.
- Built MCP server of Guidewire.
- Debugged a critical hallucination issue (agent citing non-existent policy clauses) — traced to garbled OCR chunk boundaries; fixed with a Whisper OCR + EasyOCR + Tesseract hybrid pipeline and confidence-threshold filtering

Tech Stack: *FastAPI, Python, Agno, RAG, RAGAS, LangChain, FastMCP , PgVector, Docker, LLMWhisper, Embeddings, OpenTelemetry*

Clinical Trial Management System (CTMS)

- Designed and developed a system to streamline planning, tracking, and management of clinical trials using microservices architecture.
- Integrated Keycloak for user authentication, token exchange, and permissions management.
- Integrated with EDC systems (ETMF and ISF) for secure document storage and trial data collection; implemented version control for trial protocols ensuring compliance.

Tech Stack: *FastAPI, Python, Keycloak, Microservices, PostgreSQL, Docker*

EDUCATION

Bachelor of Technology *Jun 2019 – Apr 2022*

DMS SVH College of Engineering, Machilipatnam | CGPA: 7.58/10

CERTIFICATIONS & ACHIEVEMENTS

- Extra Miler Award – Recognised for exceptional dedication at Smart IMS (2024).
- Difference Maker Award – Recognised for outstanding impact and contributions at Smart IMS, December 2025.